# Technical Proposal

## Cost-Aware Design of Seawater-Resistant Copper Alloy Powders for Blown-Powder Directed Energy Deposition

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## 1. Technical Approach

*"I earnestly believe that metallic additive manufacturing is the path to the capability and capacity you need for critical materials in the submarine industrial base."*

Matt Sermon, Executive Director, Strategic Submarines, 2023 [1]

*"Over the last year, we placed nearly 120 additive manufacturing parts into development, installed 20 of those parts, and eliminated more than 1,400 cumulative days of delay."*

Admiral Daryl Caudle, Chief of Naval Operations, 2026 [1]

### 1.1 Problem and objectives

Much of the Navy's seawater service runs on two kinds of copper alloy. Copper-nickel alloys, 90-10 (UNS C70600) and 70-30 (UNS C71500), carry seawater in piping, condensers, and heat exchangers, and nickel-aluminum bronze (NAB, UNS C95800) is cast into propellers, valves, and pump parts [2, 3, 4]. Both were developed for casting or wrought processing. Directed energy deposition (DED), in which a laser melts a stream of powder blown into a moving melt pool, can build large parts and restore worn or damaged ones without a foundry [5, 6]. DED, however, cools each layer from the melt within seconds and reheats it many times as later layers pass, so it produces microstructures these alloys were never designed for. In NAB, fast cooling can retain a martensitic phase that corrodes selectively in seawater [7, 8], and the Navy specification for NAB deposited by wire-arc DED requires the same 6-hour anneal as castings [9]. That step is costly for a large part and impractical for many repairs in place. Navy specifications for printed copper alloys use existing compositions [9, 10, 11], and designing compositions for laser deposition has only begun [12].

Designing a new composition runs through a chain of steps: a thermodynamic calculation, a powder, deposited test specimens, microscopy, and property tests (Fig. 1). The steps differ greatly in cost. A thermodynamic calculation takes minutes, a powder lot a day, a DED session days including turnaround, and a corrosion immersion weeks, because the protective film that gives copper-nickel its resistance takes weeks to months to mature [13, 14]. Experimental campaigns often send each candidate through the whole chain, spending much of the budget on full tests of candidates that a cheap early measurement could have ruled out. Computational campaigns often work as a funnel: many simulations, then a few candidates passed on to experiments. If the simulation misses a physical effect, the funnel discards good candidates before any experiment can catch the error.

The powder is a second obstacle. Single-stream blown-powder DED, the form in use today, feeds a pre-alloyed powder from one hopper. Each new composition therefore needs its own powder lot, in a size range that flows through the nozzle and that the melt pool catches [15, 16]. The lot must also fill the hopper, about 2 kg on the larger of the two systems this project will use (Section 1.8.2). Industrial gas atomizers make powder in lots of tens of kilograms or more, so new compositions are rarely tried. Our ultrasonic atomizer makes a lot of a new composition in a day from one charge of about 1 kg. Our first copper run gave over 90% powder (Section 1.5); after sieving to the DED size range, a 2 kg lot takes about three runs (225 $cm^3$ crucible), or two in the 400 $cm^3$ one. Ultrasonic atomizers have already made powder for blown-powder DED from NAB propeller grinding chips [17, 18] and from copper cable scrap [19]. Both kept the existing compositions. Making our own powder turns it from a purchase into a choice the search makes. A small lot buys cheap deposits of many compositions on a small DED system near BYU, and a full lot goes only

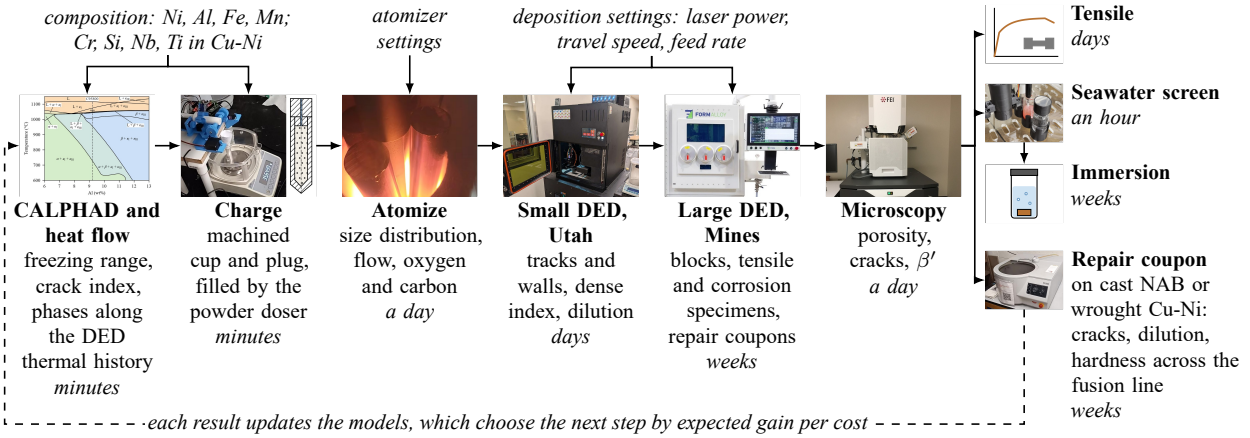



**Figure 1.** The workflow as linked models, with typical times per result in italics. Each photograph is labeled with what its step passes on, as material or as inputs to the models of later steps. DED is bought at two facilities that the search treats as two steps of different cost (detailed in Section 1.3). Utah makes small deposits from grams to about 0.5 kg of a lot. Mines takes a 2 kg lot for its hoppers, and its weeks include the extra atomizer runs and turnaround. Photographs: our laboratory, the University of Utah, the Colorado School of Mines, and the BYU Electron Microscopy Facility.

to the compositions worth building on the larger system. Making the powder also lets the search design it along with the alloy.

This project designs a copper alloy for a process: a pre-alloyed powder for single-stream blown-powder DED. The powder must deposit without cracks or pores, on itself and on the alloys it would repair, and the deposit must resist seawater without heat treatment at least as well as 70-30 copper-nickel. Getting there raises a basic question. When the steps of a materials workflow differ in cost by orders of magnitude, and results arrive at different times, how should one choose the next measurement and the next candidate? The project has five objectives:

1. Design copper alloy powders for blown-powder DED and demonstrate them from powder to deposit, repair coupon, tensile test, and seawater corrosion test.
2. Extend a cost-aware search method that learns from partial evaluations to multiple objectives: dense and crack-free deposits, strength, ductility, and resistance to corrosion in seawater.
3. Extend it to many pending experiments at once, with asynchronous results, and give it physics computed for the process: thermodynamic calculations evaluated along the thermal history of a DED deposit, and measure what that physics is worth.
4. Test whether the method's stated confidence matches what the deposits and tests show.
5. Measure how well a one-hour electrochemical screen predicts immersion tests that take weeks, so that the long tests run only where they can change a decision.

### 1.2 Copper alloys in naval seawater service

Navy specifications cover copper-nickel tube for the piping systems of surface ships and submarines [2] and NAB castings for seawater service [3]. Copper-nickel owes its resistance to a thin, adherent film of corrosion products, mainly cuprous oxide with nickel and iron, which builds up over weeks to months of exposure [13]. In one study the films on 90-10 tubes in natural seawater

were still changing after six months [14]. The iron content of the alloy changes the makeup of the film [20]. NAB is a cast alloy of copper with about 9% aluminum and 4 to 5% each of iron and nickel (by weight). It is widely used for valve bodies, pump casings, and impellers, with a large market in submarine fleets [4]. Its microstructure of $\alpha$ grains and several intermetallic $\kappa$ phases resists seawater well, but the $\alpha$ phase in the lamellar $\alpha + \kappa$ eutectoid, and any retained $\beta$ phase, corrode selectively, above all in crevices [21, 22]. Navy NAB castings therefore get a temper anneal at $1250 \pm 50$ °F (about 677 °C) for at least 6 hours, propellers excepted [3], which improves their corrosion resistance [4].

Printing changes these microstructures, and with them how the alloys corrode [23]. The microstructure of NAB follows its processing route [24], and NAB has now been printed by laser powder bed fusion [25], wire-arc deposition [26, 27], and laser DED [7, 28]. Laser-deposited NAB contains martensitic $\beta'$ along with $\alpha$ and $\kappa$ phases [7, 28] and is much stronger than cast NAB [28]. Its corrosion depends on the process. NAB from laser powder bed fusion formed stronger films than wrought NAB but resisted localized corrosion worse, and annealing reversed both [29]. Wire-arc NAB corroded in its residual $\beta$ phase [30]. Whether annealing helps depends on the starting microstructure. After 480 hours in salt solution, the same 700 °C anneal had lowered the corrosion current of cast NAB about threefold and raised that of sintered and hot isostatically pressed NAB by about as much. The authors call the differences small [31]. As-built wire-arc NAB resisted corrosion better than annealed coupons, in a test with unequal immersion times [32]. The substrate matters as well. Laser-deposited NAB on stainless steel was attacked to a depth of 3.4 $\mu$m in 21 days in salt solution where iron from the steel had diluted it, and to 1.8 $\mu$m over a nickel-chromium interlayer [33]. In welded NAB, retained $\beta'$ in the heat-affected zone corroded selectively in seawater [8], a warning for repairs. Repair work has used an NAB consumable for laser cladding of cast NAB [34]. DED repair of cast NAB has used a low-nickel copper alloy powder [35, 36], later blended in the melt pool with an aluminum alloy powder to vary the aluminum content [37].

Copper-nickel depends on the process as well. In wire-arc 70-30, the dendrite cores hold more nickel than the regions between them, and those regions corrode first [38]. Powder-bed 70-30 has been studied too [39], and its resistance depends on the laser settings [40]. At their best settings, both wire-arc and powder-bed 70-30 have matched or beaten wrought plate in electrochemical tests in salt solution [38, 40]. Laser cladding of Cu-30Ni dates to 2005 [41], and later blown-powder DED of copper-nickel has mixed copper and nickel-alloy powders in the melt pool to make graded layers and thermoelectric alloys [42, 43, 44]. The one corrosion study we found of copper-nickel deposited by blown-powder laser DED is from 2011: Cu-30Ni made by $CO_2$-laser direct metal deposition on wrought 70-30 plate [45]. Its abstract and a later review's summary of it [46] disagree on whether the deposit or the plate corroded faster. The review attributes the difference to porosity. NAVSEA has written specifications for 70-30 copper-nickel made by wire-arc DED and by laser powder bed fusion, and for NAB made by wire-arc DED. The wire-arc ones are alternatives to castings, and all three keep the existing compositions [9, 10, 11]. We found none for either alloy made by blown-powder laser DED. In 2023 a Navy official said, as Defense News reported, that for naval metals "including copper-nickel," "more research is needed" on how printing "affects the metals' fatigue and corrosion properties" [47]. A composition chosen for casting is not necessarily the best one for DED, and each composition and set of deposition settings needs its own tests.

Corrosion tests of copper alloys follow the same cost pattern as the rest of the workflow. Open-circuit potential, linear polarization, and impedance measurements in substitute ocean water [48]

take about an hour and give a corrosion rate and a measure of the surface film [49, 50]. Immersion tests with mass loss [51, 52] and cross-sections that measure the depth of selective attack take weeks. Tests in flowing seawater, which set the velocity limits used in piping design, take months. The search can therefore treat corrosion the way it treats cracking and strength: a cheap screen for many candidates and long tests for a few. The cheap screen can mislead, though. An hour-long measurement on a fresh surface sees neither the film that forms over weeks on copper-nickel nor slow selective attack in NAB. On wrought 70-30 copper-nickel in synthetic seawater, the corrosion current fell more than eightfold over the first ten days [53], and welded NAB corroded faster in natural seawater than in simulated seawater [54]. A quick measurement can also rank wrongly. In a Carderock-supported study of C63200 NAB, scanning Kelvin probe measurements in air put the copper-rich $\alpha$ phase at the highest Volta potential, yet in salt solution that phase dissolved first and the $\kappa$ phases were untouched [55]. Knowing when the screen can be trusted is part of the research.

Supply and melting add one more objective. Nickel has been on the U.S. list of critical minerals since 2022, and copper joined it in 2025 [56, 57]. Nickel also raises the melting range: 70-30 copper-nickel melts between about 1,171 and 1,238 °C, 90-10 between 1,099 and 1,149 °C, and NAB between 1,043 and 1,060 °C [58]. A replacement for 70-30 copper-nickel that needs less nickel would use less of a critical mineral and would be easier to atomize and deposit. The search will carry nickel content as an objective to minimize [59] where the corrosion trade-offs allow.

### 1.3 Scientific concepts

The search method treats the manufacturing workflow as a function network [60]. Each step gets its own statistical model (a Gaussian process), which predicts the outputs of that step and how uncertain they are. The models are linked in the same order as the steps (Fig. 1), so the outputs of one step are inputs to the next and uncertainty carries through the whole network. Because each step has its own model, the network learns from partial evaluations, and a candidate can stop at any step. Buathong et al. added to this network a cost-aware acquisition function, the rule that chooses the next experiment [61]. For each possible experiment, the rule estimates how much the result is expected to improve the search, divides that by the cost of the experiment, and runs the best one. In this project the deposition step is itself split in two by cost. A small DED system at the University of Utah, 45 miles from BYU, deposits tracks, walls, and small blocks from grams to half a kilogram of powder on a day trip. A larger one at the Colorado School of Mines, 500 miles away, builds tensile, corrosion, and repair specimens from 2 kg lots, with results in weeks. Our rule treats the two as linked steps with their own costs and pays for the larger one only where the smaller one shows that a composition could change the outcome (Section 1.8.2).

The physics comes first, in the manner of integrated computational materials engineering (ICME) [62, 63, 64, 65]: calculations decide where experiments go, and experiments correct the calculations. Calculation of phase diagrams (CALPHAD) in Thermo-Calc and pycalphad [66, 67] gives the liquidus and freezing range of each composition and an index of how prone it is to cracking as it solidifies [68]. It also gives how nickel, iron, and aluminum divide between the dendrite cores and the last liquid to freeze, and which phases are stable at each temperature, including the $\beta$ and $\kappa$ phases of the bronzes. Crack indices of this kind have been checked against printed aluminum alloys, steels, and nickel superalloys, but not copper alloys [69]. DED adds a thermal history. We model it with a moving point source of heat [70, 71], using the temperature-dependent properties of copper-nickel and NAB measured at NSWC Carderock [72, 73, 74, 75], calibrated to

measured melt-pool sizes. The temperature dependence matters: in a thermo-mechanical model of laser hot-wire NAB deposition from Penn State and Carderock, constant room-temperature properties changed the predicted distortion, at times reversing its direction [76]. For each set of deposition settings, the model gives the cooling rate through solidification and the peak temperatures that later passes impose on each layer. The CALPHAD calculations are then evaluated at those temperatures, to estimate which phases can form as the deposit solidifies and which can form or dissolve as it is reheated. These outputs enter the statistical models as inputs with learned weights, so the search can discount any of them that proves uninformative. The calculations run continuously on BYU high-performance computing, and each new result is available to the next decision.

### 1.4 Comparison with other approaches

To our knowledge, no cost-aware search over linked models that can stop a candidate at any step has been extended to multiple objectives or to asynchronous results, and none has been demonstrated on a physical alloy workflow. Published tests of the linked-model methods use test functions and simulators [60, 61]. Multi-stage Bayesian optimization for self-driving laboratories was tested on synthetic and past data with one objective [77]. Multi-information-source alloy optimization treats measurements of unequal cost as separate sources, not linked steps, even with asynchronous results, in computational campaigns [78, 79, 80]. Standard multi-objective Bayesian optimization handles competing properties and several experiments at once [81, 82, 83] but treats the whole chain as one costly experiment. Closed-loop campaigns on printed alloys use one model and a fixed set of measurements [84]. A search guided by model uncertainty has chosen multi-principal element alloys deposited by DED, scored on hardness [85], and another study combined a cheap analytical heat-flow model with costlier finite-element simulations to map DED settings [86]. Neither linked the steps nor their costs.

Alloys designed for printing by computation, such as an oxide-dispersion-strengthened NiCoCr alloy [87] and high-strength aluminum alloys [88, 89], were designed and then printed as single compositions. A nickel superalloy has been designed with a neural network for laser repair [90]. Machine learning has refined powder-bed settings for 70-30 copper-nickel at the Naval Postgraduate School [91], and Bayesian optimization has tuned a gas atomizer [92]. Both tune the process for one fixed alloy. Printed copper alloys for seawater have used the cast or wrought compositions [9, 10, 11], and their corrosion studies characterize one alloy and one process at a time [29, 30, 38, 40]. A 2026 study adjusted the composition of NAB for laser additive manufacturing to improve its as-deposited strength and corrosion resistance [12], and adding 0.1% titanium to wire-arc NAB lowered its corrosion current by 16% [93]. Both show the payoff of designing for the process.

Autonomous and high-throughput corrosion work has used a scanning droplet cell to screen coatings and has run fast screens before slower, more thorough tests [94, 95], but without deciding, candidate by candidate, when a long test is worth its cost. An ONR-supported perspective on autonomous laboratories built around additive manufacturing sets out a ten-year road map and takes as its worked example a naval brass deposited by DED, screened by a surrogate test before erosion-corrosion tests [96]. The workflow proposed here is a working instance of that idea: real step costs, DED bought by the session, and audited screens. This project combines the pieces: the linked network, the cost of each step, physics computed for the process, multiple objectives including corrosion, and asynchronous results. It then tests the combination on real deposits.

### 1.5 Preliminary results

**Powder to part.** Since the lab started at BYU in December 2025, we have set up a CALPHAD workflow on BYU high-performance computing, installed the ultrasonic atomizer, and completed a powder-to-part demonstration on a baseline aluminum alloy. Atomized powder, laser powder bed fusion builds, scanning electron microscopy (SEM) and energy-dispersive X-ray spectroscopy (EDS) results, and tensile data are in hand. The induction module of the atomizer also melts copper alloys. It reaches 1,400 °C and is typically run up to 1,300 °C. NAB and 90-10 copper-nickel melt about 250 and 150 °C below 1,300 °C, and 70-30 copper-nickel within about 60 °C of it. On 27 October 2026 we atomized our first copper-alloy lot (Fig. 3), 1,080 g of the NAB 630 bronze (UNS C63000, about 10% Al, 5% Ni, 3% Fe), at 1,300 °C on a tungsten-coated carbon-fiber plate, with an estimated 97% yield. In SEM images of 5,937 particles the powder is spherical (median circularity 0.99), with a volume-weighted median diameter of 104 $\mu$m and 70% of the volume in the 50 to 200 $\mu$m cut fed to DED. EDS at 20 kV gives $9 \pm 2$% Al, 3.4% Fe, and 5.3% Ni, within the C63000 limits, and no tungsten from the plate.

**Powder doser.** We have built an open-source auger powder doser that weighs each dose on an analytical balance (0.1 mg). A Bayesian optimization campaign tuned its eight settings unattended in under 2.5 hours. With the recommended settings it dispenses 0.5 g of salt in under two minutes, within 3 mg, where hand-tuned settings took three to six minutes. In October 2026 it began dosing atomizer charges, Al 4047 powder into aluminum cups for a composition series. The 15 wt% cup ended at 6.5676 g against a 6.5674 g target. Across twelve powders, the feed per auger turn fell as the reported angle of repose rose (Spearman $\rho = -0.84$), so its dosing record may also serve as a cheap reading of how a lot flows. Task 2 uses it the same way, for the powdered additions to each copper charge.

**CALPHAD for copper alloys.** The CALPHAD step will run primarily in Thermo-Calc (TC-Python) [66], which already runs for our aluminum work on BYU high-performance computing. The laboratory acquires its copper database [97] and maintains the software separately; no software cost is charged to the project. Open calculations supplement it: pycalphad with public databases, and machine-learning interatomic potentials adapted to phase diagrams [98]. We already run the step on copper in pycalphad [67] with an open database that covers copper, nickel, aluminum, iron, and manganese [99]. It puts the equilibrium solidus and liquidus of 70-30 copper-nickel at 1,171 and 1,229 °C, against handbook values of 1,171 and 1,238 °C [58], and those of 90-10 within 18 °C (Fig. 2). Each composition takes about one to two seconds on one processor, and a 260-point map of the nickel and aluminum ranges took 319 seconds. The map shows the liquidus passing the 1,250 °C cap of Table 1 only where nickel and aluminum are both high. For NAB the database places the solidus within 7 °C of the handbook value and predicts that $\beta$ is gone below about 730 °C at equilibrium. Any $\beta$ left in a deposit is therefore set by the cooling rate, which the process-aware descriptors of Advance 3 give the search. The database puts the NAB liquidus 51 °C too high, though, so the open calculations will serve as a check on the Thermo-Calc copper database [97].

**Deposition.** Mines has melted copper-nickel powder spread in grooves, the pre-placed tracks that start Task 2 (Facilities, Fig. 3c). At Utah, the Fang group installed its MX-Lab to develop alloys, feeding up to six powders at once [100].

**Simulated search.** We also simulated the search on a synthetic version of this workflow (Figs. 4 and 5), in which each screen misses an effect. Screening the strongest alloys that passed the crack

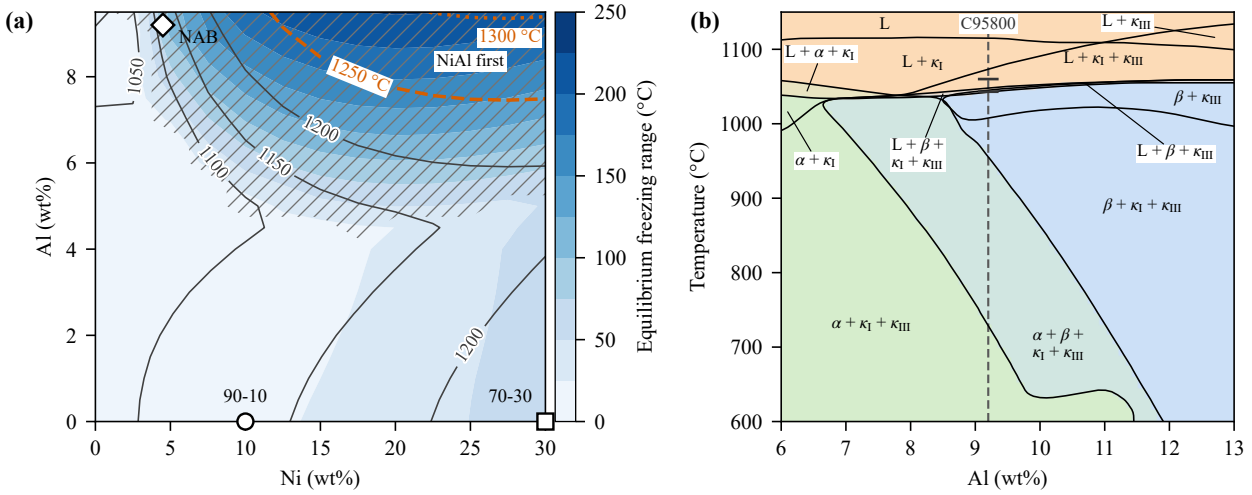



**Figure 2.** Our pycalphad calculations with an open database, mc_ni from MatCalc [99], which covers Cu-Ni-Al-Fe-Mn but was built for nickel alloys. (a) Equilibrium freezing range (shading, 25 °C steps) and liquidus (lines, °C) over the nickel and aluminum ranges of Table 1, with 1 wt% iron. Orange lines mark the 1,250 °C liquidus cap and the 1,300 °C at which the atomizer typically runs. Symbols mark reference alloys. Where hatched, NiAl forms first and the liquidus is an upper bound. (b) Vertical section through C95800 (Cu-$x$Al-4.5Ni-4Fe-1.2Mn, wt%), dashed, with its handbook solidus and liquidus as ticks [58]. Tints mark where liquid (orange), $\beta$ (blue) or $\alpha$ (green) is present. $\kappa_{\mathrm{I}}$ is iron-rich and $\kappa_{\mathrm{III}}$ NiAl-type.

screen chose aluminum bronzes that fail immersion. Searches that model the linked steps reached 73% of the optimum after $100,000 by running every candidate through the full chain and 72% by also redepositing earlier lots, against 14% for standard Bayesian optimization. The cost-aware method reached 59%: it bought about 40 large deposits per campaign but tested few. The simulation is hand-built and noise-free, with one objective and one experiment at a time, so it does not test the advances proposed here. It does show that modeling the linked steps pays off when the physics screens are incomplete, and that choosing the step needs more than the published one-step rule. A version built from a CALPHAD grid and immersion rates fitted to published same-test data gave the same order: 89% for the linked-model search with lot reuse, against 5% for standard Bayesian optimization.

### 1.6 Method advances

**Advance 1: multi-objective optimization.** The published cost-aware method optimizes a single objective. A copper alloy for DED repair must balance deposit quality (dense, crack-free, and sound at the fusion line with the part it repairs), strength, ductility, and corrosion resistance in seawater, with no weighting known in advance. We will extend the cost-aware rule, a knowledge gradient per unit cost, to multiple objectives. It will use the hypervolume knowledge gradient: the expected gain in the hypervolume of the best trade-offs the updated models would recommend, after each possible observation, divided by its cost [105]. The difficulty is that each cheap measurement informs some objectives strongly and others weakly. Single tracks show cracking and dilution well but say little about strength or corrosion, and a polarization measurement says much about corrosion rate and nothing about strength. The rule has to account for that through the network. We will derive the rule, check it against brute-force estimates on small problems, and time it on BYU high-performance

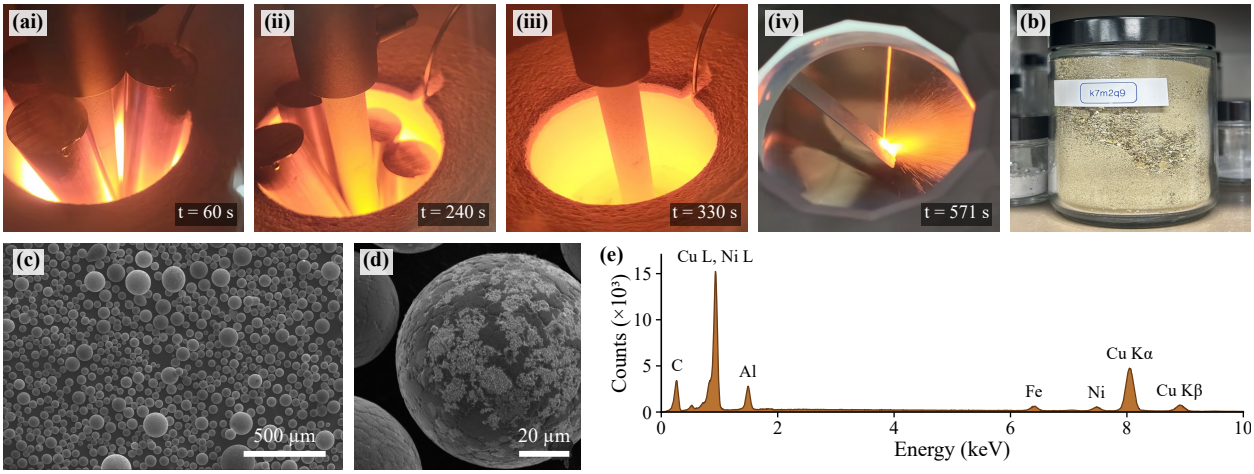



**Figure 3.** Our first copper-alloy powder, NAB (C63000), 27 October 2026. (a) The run, through the furnace port and the chamber window: (i) the charge in the crucible, (ii) melting, (iii) the inductively stirred melt around the stopper rod, (iv) the stream on the ultrasonic plate. (b) The lot. (c) SEM image of the powder and (d) one particle. (e) EDS spectrum at 20 kV; the carbon peak is from the mounting tape.

computing.

**Advance 2: pending experiments and asynchronous results.** The published method chooses one experiment at a time and waits for its result. In this workflow a DED session deposits many specimens, microscopy results arrive days later, immersion tests run for weeks, and thermodynamic calculations finish in minutes. We will extend the rule to batches of experiments with asynchronous results, so that new experiments are chosen while others are still pending. Pending experiments will enter the models as "fantasies," outcomes drawn from the current predictions, a standard way to avoid choosing the same experiment twice [106, 80]. We will also test how stable the choices are when a late result contradicts early ones. For example, a block that looked dense on its surface may show cracks under the microscope, or an alloy that passed the polarization screen may show selective attack after a month of immersion.

**Advance 3: physics computed for the process.** Standard CALPHAD inputs describe equilibrium, or solidification without diffusion in the solid, and ignore the thermal history of the process [107]. We will add inputs computed along the DED thermal history (Section 1.3): cooling rate through solidification, the peak temperatures of reheating, and the phases that CALPHAD allows at those temperatures. Hypothesis H3 tests whether these inputs make the models' predictions more reliable than equilibrium inputs, and whether equilibrium inputs beat data alone. The answer measures what incomplete, process-aware physics is worth inside a search, which sets how much simulation a campaign should buy before it buys experiments.

**Advance 4: a physical workflow from powder to repair.** We will demonstrate the advances on a physical workflow (Fig. 1) in which the search chooses the composition, the atomizer settings, and the deposition settings, and decides at each step whether a candidate goes on. After microscopy, specimens go to tensile tests or to the corrosion screen, and selected candidates to immersion. Candidates that deposit well get a repair coupon on cast NAB or wrought copper-nickel plate, checked

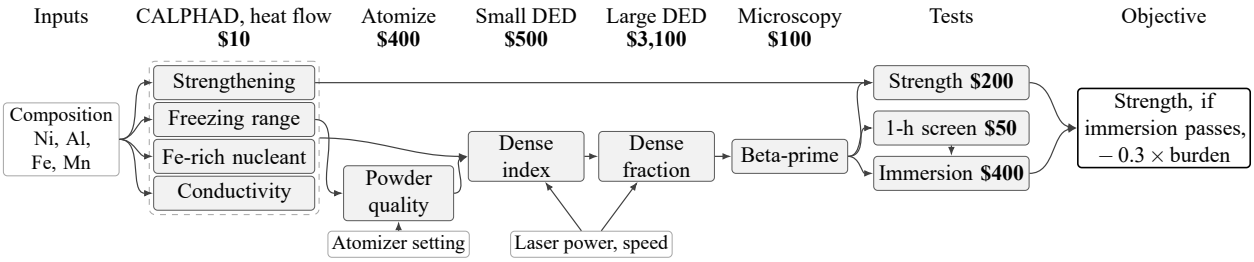



**Figure 4.** Our synthetic workflow used for Fig. 5: a synthetic function for each box and a rough all-in cost for each step ($1,500 for a repeat session at Mines). Burden is a penalty on nickel, aluminum, and manganese. Importantly, specific gaps based on literature between lower- and higher-fidelity measurements are introduced to better mimic a real-world campaign. The crack screen misses the iron-rich nucleant, as freezing-range crack indices miss nucleants that stop cracking [101, 102]. The 1-hour screen misses the attack of beta-prime, which develops over weeks [21, 22]. Utah's lower laser power leaves more lack of fusion in high-conductivity, copper-rich alloys [103, 104].

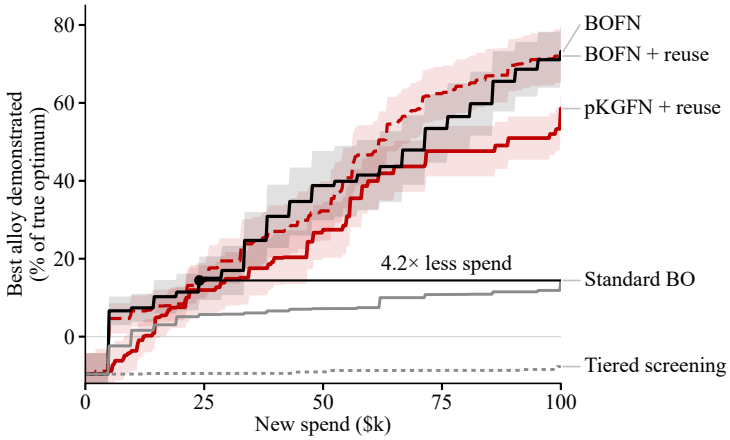



**Figure 5.** Our best result found versus spend on the workflow of Fig. 4 (means of 20 campaigns, ±1 standard error). Modeling the linked steps (BOFN [60]) matches standard BO's $100,000 result with 4.2 times less spend and reaches 73% of the optimum at $100,000, against 14%. The cost-aware pKGFN [61] needs 3.4 times less and reaches 59%. Red searches choose the next step; black and gray ones follow a fixed order. Reuse redeposits earlier lots, as pKGFN always may. Tiered screening never passes immersion.

for cracks, dilution, and hardness across the fusion line, after pre-placed tracks on them at Utah. It has real scatter between deposits, outputs measured on micrographs, and inputs that enter late.

**Advance 5: checking what the screens discard.** A cheap screen saves money only if it does not discard good candidates, and Section 1.2 shows that a short corrosion screen can also pass bad ones. Each round, the search will send a small random share of the candidates it would stop, about one in ten, on to the next step anyway. A few candidates just either side of a screen's cutoff go on as well, where its errors matter most. The audited candidates measure how often each screen is wrong, and they keep the models from learning only from candidates that the screens favored. Each year we will report, with confidence intervals, how often the one-hour corrosion screen ranks candidates as the 30-day immersion does and how often it rejects one that immersion would have passed.

### 1.7 Hypotheses and rejection criteria

**H1 (cost).** The cost-aware, multi-objective search that handles pending experiments will reach a target set of trade-offs at lower total cost than standard multi-objective Bayesian optimization, which carries every candidate through the full chain. H1 will be rejected if, across matched campaigns with the same budget, the median cost to reach the target is not lower than that of the full-chain baseline.

**H2 (calibration).** The predicted probabilities that a candidate will deposit dense and crack-free, fuse to the repair substrate without cracks, and stay close to its intended composition will match the observed rates on held-out deposits. H2 will be rejected if observed success rates differ from predicted probabilities by more than 10 percentage points.

**H3 (process-aware physics).** CALPHAD outputs evaluated along the DED thermal history will improve that calibration over CALPHAD outputs at equilibrium, which will in turn improve it over models trained on data alone. H3 will be rejected if the process-aware inputs do not reduce calibration error on unseen deposits compared with the equilibrium inputs.

**H4 (pending results).** Choosing new experiments while others are pending will reach the target in less calendar time than waiting for each batch to finish, at equal spend. H4 will be rejected if, in median calendar time, working around pending results is no faster than waiting.

**H5 (corrosion screen).** Corrosion rates, breakdown potentials, and impedance from the one-hour screen, together with composition and microstructure, will predict which candidates pass a 30-day immersion test. To pass, a candidate needs a mass loss no greater than that of the 70-30 copper-nickel reference and selective attack no deeper than in the cast NAB reference. H5 will be rejected if, among tested candidates (audited ones included), those that the model ranks in the top half pass no more often than those in the bottom half.

The targets, baselines, budgets, and number of matched campaigns will be fixed and posted before each test starts[TODO]. H1 and H4 are tested first on simulated and retrospective benchmarks, where many matched campaigns are affordable, and then in live campaigns, where only a few are.

### 1.8 Research plan

**Task 1: methods (Ph.D. student 1).** Derive and implement the multi-objective rule (Advance 1), the rule for pending results (Advance 2), the process-aware physics inputs (Advance 3), and the audit of screened-out candidates (Advance 5). Extend the benchmark of Fig. 4 with noise, several objectives, and delays drawn from measured turnaround times. Build retrospective benchmarks from our aluminum data and the copper data that Task 2 adds each year.

**Task 2: experimental platform (Ph.D. student 2).** We will make powder for each candidate in our laboratory. Oxygen-free copper bar is machined into cups that hold the charge. Nickel, iron, manganese, aluminum, and other additions are weighed in to 0.1 mg as pure metals or copper master alloys. Each run melts about 1.1 kg. The first copper run, 1,080 g of NAB, gave an estimated 97% powder (Section 1.5). Minor additions that come as powder are dispensed into the cup by our powder doser (Section 1.5), which weighs and logs each dose, so every lot carries a record of what went into it. The charge is melted in a graphite crucible in the induction module of the ultrasonic atomizer (AMAZEMET rePOWDER) and poured through a 0.5 mm nozzle onto a vibrating plate, where it breaks into droplets. Nickel and iron react with graphite. The crucible is therefore coated

with boron nitride, and the sealing rod, nozzle, and other parts the melt touches are ceramic or ceramic-coated. NAB has already been atomized from a graphite crucible on a rePOWDER [17]. Many alloys wet bare carbon-carbon composite poorly [108], and a bare carbon-fiber plate left fibers in NAB powder [18], so the plate is tungsten-coated carbon fiber, as in our first NAB run. The first part of each pour heats the plate until the melt wets it, and atomization is then continuous. With such plates on our model of atomizer, the manufacturer turned 80 to 90% of each charge of a tin bronze (CuSn8) into powder [108].

The median droplet size follows the wavelength of capillary waves on the plate, so it falls as the frequency to the power $-2/3$ [109]. The plates on our unit vibrate at 40 kHz, where the Lang relation gives a median near 47 $\mu$m for a copper melt. Measured powders are coarser. NAB atomized at 40 kHz had a median of 98 $\mu$m [18], about twice that estimate, and much of the CuSn8 powder made at 40 kHz on our model fell between 63 and 150 $\mu$m [108]. Both suit the DED system described below, which takes powder of about 50 to 150 $\mu$m, sometimes up to 200 $\mu$m, and is usually fed cuts of 45 to 105 or 100 to 200 $\mu$m. Published blown-powder work on bronzes used 45 to 106 and 45 to 150 $\mu$m cuts [28, 110]. Each lot is sieved into the two cuts that system is fed. We then measure its size distribution by sieve analysis and from SEM images, and its flow and apparent density with a Hall funnel [111, 112]. SEM also shows its shape, satellites, and stray plate fibers. NSL Analytical, an accredited laboratory, certifies the composition, oxygen, and carbon of every lot sent to Mines, after a full survey of the first copper lots.

Along with the composition, the search chooses the atomizer settings: the melt temperature, the pour pressures, and the plate amplitude. The share of each lot in each cut is an output of the powder step, and with the gas flows and the laser spot, the size distribution sets how much of the powder the melt pool catches [15]. The powder, not only the alloy, is thus designed for blown-powder DED.

We buy DED as machine time at two facilities, and the search treats them as two steps of different cost (Fig. 1). Neither facility is a subawardee. For the small step we use an InssTek MX-Lab in the laboratory of Professor Z. Zak Fang at the University of Utah, which his group has offered to the project. It has a 500 W fiber laser, 150 mm of travel on each axis, control of melt-pool height, and six powder feeders that can each be run separately [113, 100], of which we use one, with pre-alloyed powder, as a single stream (Section 1.10)[TODO]. We start every composition here, spreading powder from a first lot in grooves machined in plates, including cast NAB and wrought copper-nickel plate, and melting it at several laser powers and speeds. These pre-placed tracks use a few grams of powder each and show cracking, porosity, dilution, and fusion to the repair substrates. They differ from blown-powder tracks in heat input and in how the powder reaches the melt pool, so the models treat them as a cheaper, less faithful measurement. Once its tracks are sound, a composition gets blown-powder deposits from one 0.5 kg lot. Single tracks at several laser powers, travel speeds, and powder feed rates show melt-pool size, dilution, cracking, and the share of the powder caught in the melt pool. Thin walls show cracking and heat buildup, and small blocks show density and porosity. The heat flow model is calibrated separately for each kind of track, and the audit of Advance 5 measures how often the tracks mislead.

For the large step we buy sessions on a FormAlloy X2 blown-powder laser DED system at the Colorado School of Mines, run by facility staff at its external rates for machine time and labor. A typical session, four 25 mm cubes, takes about 3 hours of machine time and 2 hours of labor. The staff plan on the melt pool catching about half the powder fed and need about 2 kg of each powder

to fill the hoppers of the X2. Since a 2 kg lot takes about three atomizer runs, only compositions whose small blocks at Utah reach 99% relative density are scaled up. Their sessions at Mines build blocks for sub-size tensile bars (ASTM E8/E8M) and corrosion coupons at the chosen settings, and repair coupons on thick cast NAB and wrought copper-nickel plate (Advance 4). Fusing the first layer to such a large heat sink needs more laser power than the MX-Lab has. The X2 has 1 kW [114]. Commercial 70-30 copper-nickel and NAB powders, deposited at both facilities, are the baselines[TODO]. The search decides which compositions go to Mines at all.

At BYU, selected deposits are examined by SEM and by electron backscatter diffraction (EBSD) for grain size, texture, and phase fractions, including martensitic $\beta'$ in the bronzes. EDS and inductively coupled plasma mass spectrometry measure their composition, since manganese and aluminum can be lost to evaporation and oxidation [107]. Replicate tracks and blocks at fixed settings measure deposit-to-deposit scatter, and image analysis scripts turn micrographs into porosity, crack length, dilution, grain size, and phase fractions with stated uncertainty.

Each year's leading composition is gas atomized at Ames National Laboratory by the powder group of Dr. Jordan A. Tiarks, an unfunded collaborator with state-of-the-art facilities and deep expertise in gas atomization. One run a year is budgeted as a purchased service, at the end of Year 1, about two-thirds of the way through Year 2, and halfway through Year 3, so every gas-atomized lot is deposited within the award. The gas-atomized lot is deposited at Mines beside the ultrasonic lot of the same composition, so the project measures what changes between the two routes. That lot also supplies the finalists' replicate and repair builds and goes to Carderock in the form a supplier would make.

Corrosion tests follow the same staging (Fig. 6). The screen runs on the liquid-handling gantry in our laboratory (CubXL, Ursa Laboratories), which is already driven by scripts, so the search can schedule it like any other step. The gantry fills small O-ring wells, 3 to 4 mm across, clamped on polished faces, with substitute ocean water [48], and a potentiostat dedicated to the loop measures open-circuit potential, linear polarization, impedance, and cyclic polarization [49, 115]. ASTM G61 is written for iron-, nickel-, and cobalt-based alloys, so the project will report how its procedure behaves on deposited copper alloys. A potentiostat lent by a BYU colleague was integrated into the gantry's automated workflow in October 2026, although the screen itself has not yet been run. The budget buys a dedicated instrument with impedance capability, a Squidstat Plus. Small cells can add crevice attack at the seal and shift breakdown potentials [116], so each spot is photographed before and after, and the wells are checked against a conventional 1 $cm^2$ flat cell on the references. Several wells fit on one specimen, so a repair coupon is measured in the deposit, at the fusion line, and in the heat-affected zone of the substrate, where welded NAB was attacked selectively [8]. The gantry then pipettes the used electrolyte of each well into a tube for inductively coupled plasma mass spectrometry, which shows which elements left the surface, for example aluminum from NAB or nickel from copper-nickel.

Wrought 70-30 copper-nickel and heat-treated cast NAB are measured as references in every round. Because the gantry runs unattended, it can also re-measure specimens left immersed for days at little extra cost, a step between the one-hour screen and the 30-day immersion.

Candidates that the search selects get 30-day immersion tests in capped vials on the deck of the gantry. It draws a few microliters of each solution at about 1, 3, 10, and 30 days for the same analysis. The metal released over time then enters the models as an intermediate result between




**Figure 6.** Our corrosion screen on the gantry. Left: the gantry as it is now. Right: one spot of the planned protocol, rendered from the laboratory's animation. (i) Specimens and well lids, (ii) substitute seawater fill, (iii) electrode cap, (iv) open-circuit potential, impedance, and linear polarization (cut-away), (v) cap rinsed, (vi) the well's liquid sampled for ICP-MS, (vii) the spot imaged. Impedance runs on the requested potentiostat, and cyclic polarization on selected spots.

the one-hour screen and the final mass loss. The coupons are weighed before and after [51, 52] and sectioned to measure the depth of selective attack. Repair coupons are immersed as galvanic couples, the deposit on its cast NAB or wrought copper-nickel substrate with the fusion line exposed and the area ratio recorded, and are sectioned across the fusion line to find the deepest attack. The leading candidates get crevice and longer immersion tests[TODO]. Tests in flowing seawater, which take months, are left to Navy laboratories, such as the marine corrosion laboratories at NSWC Carderock [117], for the final candidates, with replicate coupons of each (Section 1.10).

The search space was chosen by a literature review of each candidate element, then a check of its coverage in the CALPHAD database. Fifty elements from the periodic table (Fig. 7) were reviewed one at a time, including every element that appears in a commercial copper alloy, a copper alloy or welding filler specification, a deoxidizing practice, or the atomizer's crucible and plate materials. Each review asked four questions. Does the element help seawater resistance, deposit soundness, or repair? What sets its upper limit? Can it be melted and atomized below 1,300 °C, where the atomizer typically runs, without attacking the crucible or plate? Does Thermo-Calc's copper database (TCCU7 [97]) assess its systems? Eight elements are searched from the start (Table 1), with copper as the balance at no less than 65% and a calculated liquidus no higher than 1,250 °C. From Year 2 we will also test five microalloying additions (Table 1).

Rules tied to the aluminum content keep each candidate near known metallurgy. Above 7% aluminum, nickel must at least equal iron, as it does in the nominal composition of Navy NAB castings [3]. Below 1.5% aluminum, iron stays at or below 2.3%, the upper limit of the iron-rich 70-30 grade C71640, so that it remains in solid solution [118]. Chromium and niobium are used only where aluminum is at most 3%, and silicon above 0.1% only with niobium, as in cast 70-30 copper-nickel [119]. No fixed formula limits retained $\beta$. The process-aware CALPHAD outputs (Advance 3) enter the search as descriptors, and it learns the retained phases from those measured in each deposit. The space includes the region between the two families, copper-nickel-aluminum alloys such as the high-strength MARINEL [120], where low-nickel candidates may lie. Beryllium and cadmium (toxic powder and fume), zinc (it evaporates in the melt pool and dezincifies in seawater), and the refractory metals (immiscible, and far above the melting limit) are excluded. Lead, sulfur, phosphorus, carbon, bismuth, and the other impurities are held at or below the welding-grade limits for copper-nickel [121]. All the searched elements but iron are on the 2025 U.S. list of critical minerals [57]. Nickel dominates by mass and is carried as an objective.

**Task 3: live campaigns and hypothesis tests (both students).** Run live campaigns in which the search chooses each experiment and step. Compare it on equal budgets with random selection and with standard multi-objective Bayesian optimization over the full chain. Test H2 and H3 on held-out

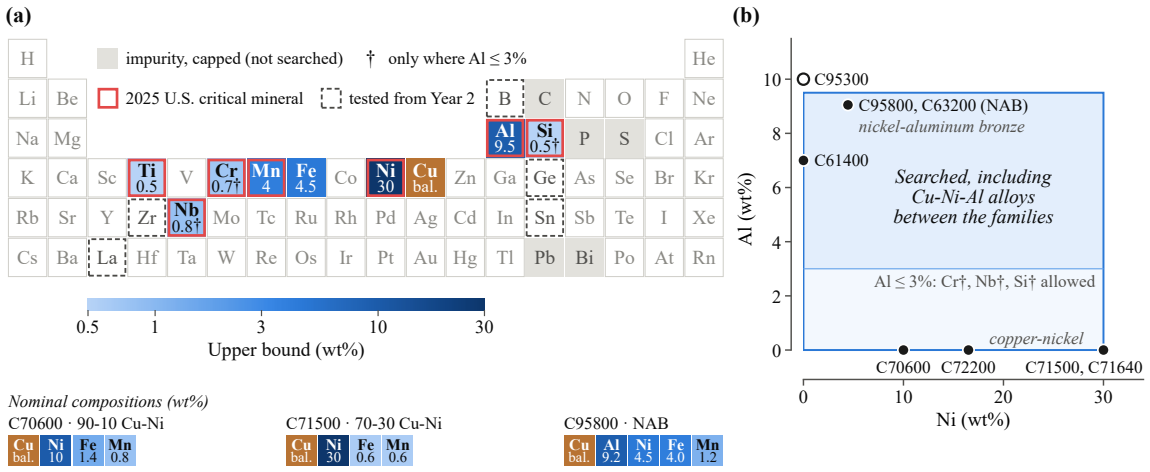



**Figure 7.** Our search space. (a) Elements searched from the start, shaded by upper bound (Table 1), with copper as the balance. Red outlines mark critical minerals on the 2025 U.S. list [57]. Gray tiles are impurities that are not searched but are held at or below the limits for copper-nickel welding filler [121]. †Added only where aluminum is at most 3%. Below, on the same scale, the nominal compositions of the three reference alloys [58]. (b) Familiar alloys at their nominal compositions in the nickel-aluminum plane. The shaded box is the searched range, which includes the copper-nickel-aluminum alloys between the two families. The †elements are allowed only below the 3% aluminum line. Open circle: outside the range.

deposits, H5 on the audited and selected candidates, and H1 and H4 in matched live pilot campaigns, confirmed on the simulated and retrospective benchmarks; the budget justification allocates each Mines session. Researchers review each batch before it runs, can veto or add candidates, and log every override. As a stretch goal, a language model will turn their written feedback into preferences the search can model [137].

**Task 4: release and transition.** Release the software, data, and models each year (Section 5). The dataset records every step and measurement for every candidate: charge preparation, atomization, sieving, storage and transport, deposition, and the mechanical and corrosion tests. Each property thus traces back to its powder lot and thermal history. The dataset follows the form of the temperature-dependent property databases that Carderock publishes [75, 72], and the PI has deposited widely used datasets before [138, 139]. Write a short guide for applying the method to other alloys, deposition processes, and test sequences, aimed at Navy laboratory and depot engineers. For each final candidate, publish its powder specification, deposition settings, and tensile, corrosion, and repair-coupon data. Share them with materials engineers at NSWC Carderock, who can decide which merit flowing-seawater tests and, beyond this project, Navy qualification testing.

### 1.9 Expected results and new knowledge

If successful, the project will produce:

1. The first cost-aware method of this kind for multiple objectives and asynchronous results.
2. A measured answer to how much a staged, cost-aware search saves over full-chain testing on a real alloy workflow.

**Table 1.** Elements searched from the start (wt%), with copper as the balance (at least 65%) and a calculated liquidus of at most 1,250 °C, and the additions tested from Year 2 (gray). †Only in alloys with at most 3% aluminum (silicon above 0.1% only with niobium).

| Element | Range | Role | Upper limit set by |
|---|---|---|---|
| Ni | 0–30 | Protective film in Cu-Ni; $\kappa$ phases in NAB [13, 21] | Liquidus near the atomizer's limit; supply (an objective) [58, 57] |
| Al | 0–9.5 | Aluminum bronze; Cu-Ni-Al alloys [4, 120] | Retained $\beta'$ martensite as deposited [28] |
| Fe | 0–4.5 | Film in Cu-Ni; $\kappa$ phases in NAB [20, 21] | Iron out of solution in Cu-Ni; coarse $\kappa$ in NAB [122, 123] |
| Mn | 0–4 | Deoxidizer; partial nickel substitute [110] | Evaporation in melting and DED [124] |
| Cr | 0–0.7† | Resistance to fast flow [125] | Cr-rich particles; crevice attack in still water [125] |
| Si | 0–0.5† | Deoxidizer; strength with Nb [126, 119] | Silicides; weld cracking [126] |
| Nb | 0–0.8† | Weldability of cast Cu-Ni [126, 119] | Undissolved Nb-rich particles [127] |
| Ti | 0–0.5 | Deoxidizer in Cu-Ni welding filler [121] | Oxide and nitride inclusions; crucible attack [128] |
| *Microalloying additions tested from Year 2* | | | |
| Zr | 0–0.15 | Grain refiner [129] | $ZrO_2$ inclusions; excess at grain boundaries [129] |
| B | 0–0.02 | Grain refiner [129] | Borides; Ni-B eutectic below the Cu-Ni solidus [130] |
| Sn | 0–1, NAB only | Grain refinement and strength in cast NAB [131] | Wider freezing range [132] |
| Ge | 0–0.2, NAB only | Erosion-corrosion (one study) [133] | Cost; export controls [134] |
| La or Ce | 0–0.1† | Resistance in salt solution [135] | Coarse rare-earth inclusions above about 0.1% [136] |

3. A measured value of process-aware physics: how much CALPHAD evaluated along the DED thermal history improves predictions over equilibrium calculations and over data alone.
4. Corrosion data for copper alloys deposited by DED, and a measured link between a one-hour electrochemical screen and immersion tests that take weeks.
5. New copper alloys designed for blown-powder DED, each with its powder specification, deposition settings, tensile properties, corrosion behavior, and repair coupons, using less nickel where the trade-offs allow, and each a candidate for Navy qualification testing.

Results that reject a hypothesis are also new knowledge. If H1 fails, the data will show which step's cost or noise erased the advantage. If H3 fails, they will show which process effects the calculations missed, and if H5 fails, which microstructural features the screen missed.

### 1.10 Future Naval Relevance

**ONR S&T Focus Areas.** The primary focus area is Manufacturing, research area Integrated Manufacturing and Sustainment Technologies. The secondary focus area is Materials and Electronics, research area Sustainable and Operationally Robust Materials. Within Code 332 the project fits the program Materials & Processes for Additive Manufacturing, whose thrusts include "tailored

AM materials and processes," "large scale AM," "additive repair," and "rapid qualification frameworks" [140]. The project serves the first three with an alloy designed for a large-scale process that also repairs, and the last with a search method that decides which test to run next and states its confidence in the outcome. It is ICME for additive manufacturing: computation first, and experiments where they change a decision. It also fits two other Code 332 programs. Corrosion Science and Corrosion Control Technologies asks for a "fundamental understanding of corrosion mechanism in additive manufactured (AM) alloys" [141], and Computer-Aided Materials Design lists high-throughput screening concepts and informatics among its thrusts [142].

**Copper alloys, castings, and repair.** The slow production of large castings "has historically gated progress and throughput for both the Columbia and Virginia submarine programs" [1], and the Navy is replacing castings with printed parts. In August 2026 it set out "a streamlined, repeatable technical pathway to substitute legacy cast and wrought parts with 3D-printed metallic materials" [143]. A printed copper-nickel deck drain, made "to provide an alternative to traditional copper-nickel castings," has been installed on a new-construction submarine [144, 145], and the Navy Maritime Industrial Base program has made copper-nickel ship-service ball valves by wire-arc deposition [146]. These parts use the cast compositions, as does ONR-funded research on wire-arc 70-30 copper-nickel [38]. Suppose a copper alloy designed for blown-powder DED resisted seawater as deposited. A shipyard or depot could then build a part that now waits on a casting, or restore a worn part in place without the heat treatment the cast alloys need. ONR's corrosion program asks for this kind of work: "modify the chemistry of feed stock materials to enhance AM processing and corrosion resistance of additive-manufactured alloys" [141].

**The longer horizon.** The project designs for single-stream, pre-alloyed powder, which existing DED systems can use now. Blended powders separate in the feed stream and are caught by the melt pool at different rates, which shifts the deposited composition [147, 148, 42]; a pre-alloyed lot carries one measured composition into every deposit [148]. The same models can later choose blends fed from several hoppers and alloyed in the melt pool, so a depot could make a new composition without a new powder lot. The method also applies to wire-arc and laser-wire deposition and to other Navy alloys, including titanium.

**Navy engagement and people.** The Naval Surface Warfare Center (NSWC) Carderock Division is the Navy lead for additive manufacturing technology and the NAVSEA lead for welding process and material specifications [149], and it runs marine corrosion laboratories [117]. Its engineers have studied how processing sets the microstructure of NAB [24] and published the temperature-dependent properties of copper-nickel, NAB plate, and wire-arc NAB that our heat flow model uses [72, 73, 74, 75, 76]. The dataset of Task 4 will be released in the same form. The PI remotely presented autonomous experimentation for alloy discovery to Carderock on 14 October 2026, hosted by Dr. Nicholas J. Jones, to an audience of approximately 20[TODO]. Each year we will send the candidates and data to materials and corrosion engineers at Carderock and ask which flowing-seawater and repair tests the final candidates should get. We will visit in Years 2 and 3 on the trips to the ONR program review, about 15 miles away. The PI will apply to the Summer Faculty Research Program to spend a summer at Carderock[TODO]. The project will train two Ph.D. students and undergraduates in powder making, deposition, characterization, corrosion testing, and autonomous experimentation. We will urge them to apply to the Naval Research Enterprise Internship Program and the SMART and NDSEG programs, which the NOFO names as YIP goals.

### 1.11 Project Schedule and Milestones

Table 2 lists the schedule by year from the award start date of 1 April 2027.

**Table 2.** Schedule and milestones (Y = project year, starting 1 April 2027).

| Year | Work | Milestone (month) |
|---|---|---|
| Y1 | Derive the multi-objective rule; build simulated and retrospective benchmarks; build the process-aware physics inputs; first copper powder lots in the DED size range and their tracks at Utah; deposit the commercial baselines at both facilities and replicate them to measure scatter; set up the corrosion screen on wrought and cast references; automate image analysis. | M1: first copper powder lot sieved, checked, and melted in tracks at Utah (4).<br>M2: baseline deposits and scatter measured (7).<br>M3: corrosion screen repeatability measured (8).<br>M4: multi-objective rule passes small-problem checks (9).<br>M5: benchmarks frozen and posted (12). |
| Y2 | Derive the rule for pending results; first live campaign with tensile, corrosion, and repair branches; first 30-day immersions; test H2 and H3 on unseen deposits. | M6: first live campaign done (18).<br>M7: H2 and H3 tested (22).<br>M8: methods paper and software release 1 (24). |
| Y3 | Matched live campaigns against the full-chain baseline; test H1, H4, and H5; release software, data, and guide; share powder, deposition, corrosion, and repair data with Navy engineers. | M9: matched campaigns done (32).<br>M10: H1, H4, and H5 tested (34).<br>M11: final release and report (36). |

### 1.12 Goals and Benchmarks

Table 3 lists each goal and the benchmark used to measure progress. Progress will be reported at each ONR program review.

### 1.13 Gold Standard Science

Executive Order 14303 lists nine tenets of Gold Standard Science [150]: reproducible, transparent, communicative of error and uncertainty, collaborative and interdisciplinary, and skeptical of its findings and assumptions. Such science is also structured for falsifiability of hypotheses, subject to unbiased peer review, accepting of negative results as positive outcomes, and without conflicts of interest. The plan follows each point. Each hypothesis above has a stated rejection criterion, and we fix the targets and baselines before each test. Every prediction carries an uncertainty, and the calibration tests (H2, H3) check whether that uncertainty is honest. The random audit of Advance 5 tests the assumptions behind each screen. Replicate deposits and reference specimens measure

**Table 3.** Goals and the benchmarks that measure them.

| Goal | Benchmark |
|---|---|
| Multi-objective cost-aware rule | Matches brute-force estimates on small problems; finds a larger set of trade-offs per dollar than a full-chain search on the simulated benchmark. |
| Rule for pending results | Same final quality as the one-at-a-time rule in less calendar time on the simulated benchmark with measured delays (H4). |
| Calibrated predictions | Observed success rates within 10 percentage points of predicted probabilities on unseen deposits (H2). |
| Process-aware physics | Lower calibration error with thermal-history inputs than with equilibrium inputs, and with equilibrium inputs than without (H3). |
| Powder for blown-powder DED | Every deposited composition made in-house, sieved to the DED size range, with its size distribution, flow, and oxygen and carbon content recorded. |
| Corrosion screen | Top-ranked candidates pass the 30-day immersion test more often than bottom-ranked ones (H5). |
| Lower cost on real deposits | Lower median cost to reach the target than the full-chain baseline in matched live campaigns (H1). |
| New alloys | At least three new compositions, one a year, deposited dense and crack-free, with tensile, corrosion, and repair-coupon data, beyond the commercial baselines. |
| Open release | Software, data, and models posted each year with tests and documentation. |

scatter so that it is not mistaken for a real effect. Code, data, and models are released each year, so others can rerun every analysis. Results that reject a hypothesis will be published like any other result. The PI's record supports this: he co-authored a review of self-driving laboratories and led open benchmark and software projects [138, 139].

## 2. Management Approach

In this single-PI project, with no co-PIs or subawards, the PI will lead the method development, set the campaign plans, and supervise all students. Ph.D. student 1 will lead Task 1 (methods, physics inputs, benchmarks, software). Ph.D. student 2 will lead Task 2 (powder, deposits, microscopy, tensile and corrosion tests). An undergraduate researcher will help make and check powder, prepare specimens, and run the corrosion screen. Both Ph.D. students share Task 3, so the methods are tested on the data they will face.

DED time is bought from the University of Utah and the Colorado School of Mines, and any outside tests from a commercial laboratory. None is a subaward. A BYU student runs the Utah sessions after training by the Utah group, and staff at Mines run each session there, both to the settings the search chooses. Dr. Jordan A. Tiarks of Ames National Laboratory is an unfunded collaborator for the gas atomization runs. Fang and Webster are unfunded collaborators too, and all three are listed under current and pending support. Our budget buys about twelve half-day sessions a year at Utah,

with several compositions to a session. It also buys 10, 13, and 10 sessions at Mines in Years 1 to 3 and one gas atomization run a year[TODO]. All other work is at BYU: the PI's atomizer laboratory, the BYU Electron Microscopy Facility (SEM, EDS, EBSD), and BYU high-performance computing. The team meets weekly to review new results and choose the next experiments. A shared database records every step, so the search always sees the latest results.

**Risks and responses.** If the multi-objective rule is too slow to compute, we will use a cheaper approximation and measure the loss on the simulated benchmark. If deposit scatter is large, we will add replicates and enter the scatter in the models directly. If a composition cannot be atomized, for example because it melts too close to the limit of the module, the search records the failure as a result, since predicting which compositions can be made is part of the task. If the coating of a plate wears and the melt stops wetting it, we replace it with a fresh tungsten-coated carbon-fiber plate, as in our first copper run. If melts that contain nickel or iron still pick up carbon, which the carbon analysis of each lot will show, we will recoat the crucible before every run instead of every few. If a lot flows poorly, the search sees that at the powder step and can change the atomizer settings before paying for a deposit. The pre-placed tracks on cast NAB and copper-nickel plate show early whether the first layer fuses to thick plate. If it does not, we will preheat the repair coupons or make them thinner. If 500 W at Utah cannot melt a composition well, its tracks move to Mines at higher cost, and the search learns which compositions need the larger system. If making 2 kg lots slows the campaigns, the Utah step will carry more of the screening, and fewer compositions will be scaled up. If either facility becomes unavailable, the other takes its work at its known cost. If DED turnaround is slow, the rule for pending results is designed for that case. If the electrochemical screen predicts the immersion tests poorly (H5 rejected), the search will rely more on immersion, at higher cost, and the audited candidates will show which features the screen missed.

## 3. Principal Investigator Qualifications

Sterling G. Baird received a Ph.D. in Materials Science and Engineering from the University of Utah in August 2023. He has been an Assistant Professor of Mechanical Engineering at BYU since December 2025, his first tenure-track position. Before BYU, he was Director of Training and Programs, then Principal Investigator and Staff Scientist, at the Acceleration Consortium, University of Toronto. There he built its development laboratory with about $600,000 of equipment and helped direct the organization's growth from 6 to 60 full-time employees (35 staff scientists) in three years. He has published 30 peer-reviewed journal articles, 12 as first author (h-index 14, Google Scholar).

The PI combines Bayesian optimization with laboratory hardware. He developed Honegumi, which helps experimentalists set up Bayesian optimization [151], consulted for Meta on the version 1.0 release of the Ax platform, and organized an international Bayesian optimization hackathon [152]. He co-wrote the proposals for a Fehrmann Alloys industry project on designing additively manufactured aluminum alloys with ICME and multi-task, multi-objective Bayesian optimization and for an NSF EAGER award on extracting materials data with large language models. He wrote the structural materials section of the *Chemical Reviews* survey of self-driving laboratories [153] and led a review of low-cost ones [154]. He also designed a semi-automated induction furnace for temperatures up to 2500 °C [155]. His electrochemistry experience includes lithium-oxygen batteries at NASA Langley Research Center, hydrogen embrittlement studies, and electrochemical metallographic preparation. At BYU he supervises about 20 undergraduates and one M.S. student and set up the CALPHAD workflow, atomizer, powder doser, and powder-to-part demonstration.

## Principal Investigator's Curriculum Vitae

TODO

# Letter of Support

TODO

## 4. Responsibility

**Resources.** Our laboratory at BYU holds the ultrasonic atomizer, the powder doser, the liquid-handling gantry, and the equipment for powder handling and specimen preparation[TODO]. The loaned potentiostat is integrated into the gantry's automated workflow and serves until the one item of equipment requested, a dedicated potentiostat with impedance capability, arrives. The BYU Electron Microscopy Facility provides SEM, EDS, and EBSD, and BYU provides high-performance computing. We buy DED time at Utah and Mines, as described under Task 2. Tensile tests use the Instron load frames in the materials testing laboratory of the Department of Mechanical Engineering[TODO].

**Ability to comply.** The current commitments of the PI are teaching, start-up research, and the pending requests listed under current and pending support. The effort proposed here fits within those commitments.

**Performance history.** This would be the PI's first federal award since he began as a PI at BYU. BYU Research Administration will manage it.

**Integrity and business ethics.** The PI has no findings of research misconduct and follows BYU policies on research integrity and conflicts of interest.[TODO]

**Qualifications and eligibility.** The PI received his Ph.D. after 1 January 2019, holds his first full-time tenure-track position, and has not received an ONR YIP award. He is a U.S. citizen. BYU is registered in SAM.gov (UEI JWSYC7RUMJD1, CAGE 52534)[TODO].

**Organization and controls.** BYU Research Administration manages federal awards, with accounting, purchasing, and property control systems[TODO]. Laboratory safety follows BYU Risk Management and Safety procedures. The atomizer stands in a steel-panelled enclosure built inside the laboratory in 2026 with the college safety officer (Fig. 8): its own exhaust, argon supply, and dehumidifier, a Class D extinguisher, an explosion-proof vacuum, and respirators.

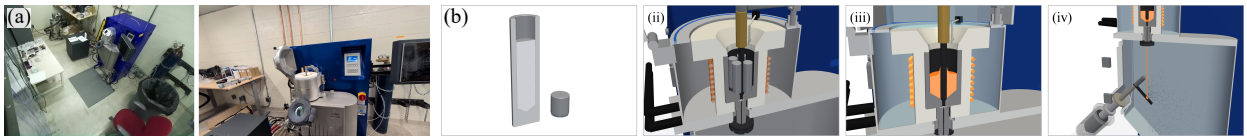



**Figure 8.** (a) The humidity-controlled, ventilated powder enclosure in the PI's laboratory that houses the atomizer, and the atomizer with its induction module open (28 September 2026). (b) One run, rendered from a CAD model of the atomizer: (i) cup and plug, (ii) cups in the crucible, (iii) melting and stirring, (iv) pouring onto the ultrasonic plate.

The handling rules were set for aluminum powders, the most hazardous case in the Bureau of Mines series [156]. They came out of conversations through 2026 with Ames National Laboratory (Iver Anderson and Jordan Tiarks), North Carolina State University (Tim Horn), the Johns Hopkins group that runs the same atomizer, the manufacturer, and an aluminum powder producer. Copper alloy powders are far less reactive, and the same controls apply. Nickel-bearing powders are handled in glove bags with respiratory protection. The first copper alloy powder is tested for ignition energy, explosion severity, and transport class before any lot ships, and ignition energy is retested on each gas-atomized lot. Corrosion tests use salt solutions. The dilute acids used to clean immersion coupons [52] are handled in a fume hood under BYU chemical safety procedures.

## 5. Data Management Plan

**Types of data, software, and other materials.** We will produce thermodynamic and heat flow calculation results, and atomizer settings with records of powder size, flow, density, oxygen, and carbon. Settings and session logs cover the pre-placed tracks and DED deposits. Microscopy gives optical and electron micrographs, EDS spectra, and EBSD maps. Mechanical tests give tensile and hardness data, and corrosion tests give polarization and impedance data, immersion mass losses, and cross-sections of corrosion attack. The software products are the search software, benchmark problems, and trained models.

**How the data will be acquired.** Calculations run on BYU high-performance computing. We make powder in our laboratory. DED sessions are bought as a service (Section 2). Microscopy, corrosion screens, immersion tests, and mechanical tests are done at BYU.

**Time and location.** Data are collected at BYU and at the two DED facilities (University of Utah and Colorado School of Mines) from April 2027 to March 2030.

**Processing.** Scripts kept under version control turn raw images and test records into derived quantities (porosity, crack length, dilution, grain size, phase fractions, strength, corrosion rate, attack depth) with stated uncertainty. Each derived value links to its raw files.

**File formats and naming.** Open formats: CSV and JSON for tables, TIFF for images, and the instruments' native formats alongside open exports. Each sample receives an identifier that encodes powder lot, deposition session, and specimen, used in every file name.

**Quality assurance and control.** Replicate deposits measure scatter. We measure reference specimens of wrought 70-30 copper-nickel and cast NAB in every round of corrosion screening. Scripts are tested, and the same benchmarks are rerun before each software release.

**Existing data.** The retrospective benchmarks use our own aluminum data from 2026, with its origin recorded in the metadata.

**Standards.** Metadata will record every step from charge preparation through atomization, sieving, storage, transport, deposition, and testing. For each step it will hold the composition, powder properties, process and instrument settings, dates, operators, and the ASTM method behind each test. The schema is informed by the one used by NIST AM Bench [157], which covers builds, specimens, and measurements, and we extend it to powder making and corrosion tests.

**Preservation.** Data and software will be deposited in public repositories at each annual release and kept for at least 10 years. Data go to Zenodo or figshare with a DOI, as the PI's earlier datasets did [138], and the property databases of the finalists also go to Materials Commons, where Carderock released the raw files of its HSLA-80 database [158]. Code goes to GitHub.

**Balance of value and cost.** We will keep raw session logs and micrographs in full. Very large intermediate files that scripts can regenerate will not be archived.

**Release restrictions.** Results are expected to be fundamental research. Any data that cannot be cleared for public release under DoD Instruction 5230.09 will not be made public.

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**Open TODOs for Sterling**

Not part of the submission or the page count; each is marked TODO in the text, and the list is rebuilt from the source by `make_todo_list.py`.

1. **H5 (corrosion screen).** confirm a public pre-registration site, e.g., OSF
2. **Task 2: experimental platform (Ph.D. student 2).** confirm with Fang: the laser power (300 or 500 W), the rate, the smallest powder charge per feeder, the argon chamber and any preheat, and that a BYU student runs it after training
3. **Task 2: experimental platform (Ph.D. student 2).** confirm commercial powder sources, at least 2 kg of each; Mines can also supply powder at cost
4. **Task 2: experimental platform (Ph.D. student 2).** confirm the durations, and natural or substitute seawater
5. **Navy engagement and people.** name any Carderock contacts who agree to be listed, and confirm Dr. Jones's title and code
6. **Navy engagement and people.** confirm; the 2027 application closes 14 December 2026; Carderock's SFRP topics include sustainment, repair, and corrosion science
7. **Management Approach.** the Utah line is an allowance until Fang sets a rate
8. **Principal Investigator's Curriculum Vitae.** insert the full CV PDF here.
9. **Letter of Support.** insert the signed letter from the BYU official here as a PDF page (draft in support-letter-draft.md; it names aluminum, so revise it for this version).
10. **Resources.** list the main items bought with start-up funds
11. **Resources.** confirm
12. **Integrity and business ethics.** confirm.
13. **Qualifications and eligibility.** BYU's SAM registration is listed as ending 4 March 2027, before the award start; confirm that RAO will renew it
14. **Organization and controls.** confirm wording with BYU Research Administration